

NITHIN BALAMURUGAN

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Experience

Data Engineer

Sept. 2026 – Dec. 2026

TD

Toronto, ON

- Engineered Azure **ETL pipelines** with **PySpark** on Databricks, processing 2,500–3,000 documents per ingestion cycle (**Databricks, Azure SQL, Data Lake**).
- Deployed ML models** for **data curation** and optimized embedding **infrastructure**, boosting retrieval accuracy 95%→98% while cutting provisioned compute 47%.
- Built cloud infrastructure**, CI/CD, and Datadog monitoring across **DEV/TEST/PROD**, enabling reliable deployments and faster incident triage.

Data Engineer

May 2026 – Aug. 2026

CIBC

Toronto, ON

- Resolved **100+ production incidents** via root-cause analysis, maintaining **98%+ uptime** across 4+ systems (**Azure, Databricks, SQL**).
- Managed/improved **Databricks** queries + **Azure Data Factory** pipeline reliability/workflow
- Supported **deployments** and release management through **Azure DevOps** and **Salesforce**, improving production workflows and deployment reliability.

Software Engineer

Sept. 2025 – Dec. 2025

CognitomeAI

London, ON

- Built a **Python/FastAPI** data pipeline to ingest, clean, index, and retrieve PhD and peer-reviewed papers using **NLP, embeddings, vector search, and semantic similarity**.
- Developed an interactive data visualization that clusters and ranks papers by relevance using statistical analysis and similarity scoring, supporting faster data-driven decision making.
- Built and **deployed a GPT-3.5 RAG chatbot** using Python, Docker, **AWS**, and Next.js for citation-backed research.

Research Publications/Projects

Tiny Recursive Models on ARC-AGI-1 | [arXiv:2512.11847](#) | [Research Blog](#)

- Built **reproducible data pipelines** for ARC-AGI-1 experiments, processing and evaluating **400+ tasks** at scale.
- Engineered a **1,000-sample data augmentation pipeline**, improving Pass@1 accuracy by **10.75/% points**.
- Ran **data analysis** and model **ablation** experiments, identifying puzzle-ID embeddings as a critical performance dependency.
- Benchmarked GPU throughput, latency, and memory**, comparing a 7M-parameter TRM against Llama 3 8B.

CustoMatch | *ML-based recommendation engine matching brands to retailers*

- Built a retailer-brand recommendation engine (**two-tower embeddings, pgvector**) served via **FastAPI**.
- Automated** the ML pipeline with **dbt, PySpark, and Airflow** – daily retraining and drift monitoring.

StoryOS | *3D Object Rendered Video Gen/Editing* | [Project Page](#)

- Built a full-stack knowledge graph platform (**Next.js, TypeScript, FastAPI, Neo4j**) from ingested documents and URLs.
- Designed an async ingestion pipeline (**RabbitMQ, Redis-backed Celery**) to scrape and process documents at scale.
- Integrated **OpenAI** function calling for entity extraction; deployed on **AWS** via **Vercel**.

Technical Skills

Languages: Python, C/C++, Java, TypeScript, JavaScript, SQL (PostgreSQL)

AI / ML: PyTorch, Hugging Face Transformers, XGBoost, SHAP, ONNX, OpenAI API, Depth Anything V2, DPT-MiDaS

Data & Cloud: NumPy, Pandas, AWS, Docker, Modal, Firebase Firestore, PostgreSQL

Backend & Tools: FastAPI, Flask, Node.js, Next.js, React, Git / GitHub CI/CD, Jira

CV / Media: OpenCV, FFmpeg, OCR (Tesseract, RapidOCR)

Education

The University of Western Ontario

Expected Apr. 2027

Bachelor of Computer Science

London, ON

- Relevant Coursework: Linux, ARM Assembly, Data Structures & Algorithms, Systems Programming, OOP, Database Systems, Networking